Surface Areas and Volumes

The big idea

Cones, spheres and hemispheres are all around us: ice-cream cones, footballs, bowls. This chapter gives the formulas for how much material covers them (**surface area**) and how much they hold (**volume**).

Right circular cone

A cone has radius r, height h and **slant height $l = \sqrt{(r^2 + h^2)}$**.

CONE

Curved surface area = πrl

Total surface area = $\pi r(l + r)$

Volume = $\frac{1}{3}\pi r^2 h$

A cone's volume is **one third** of a cylinder with the same base and height: three cones of water fill the cylinder.

A CONE HAS RADIUS 7 CM AND SLANT HEIGHT 10 CM. FIND ITS CURVED SURFACE AREA.

CSA = $\pi rl = (22/7) \times 7 \times 10 =$ **220 cm²**.

FIND THE VOLUME OF A CONE WITH RADIUS 6 CM AND HEIGHT 7 CM

$\frac{1}{3} \times (22/7) \times 36 \times 7 =$ **264 cm³**.

Sphere and hemisphere

SPHERE AND HEMISPHERE

Sphere: surface area = $4\pi r^2$, volume = $\frac{4}{3}\pi r^3$

Hemisphere: curved surface = $2\pi r^2$, total surface = $3\pi r^2$, volume = $\frac{2}{3}\pi r^3$

A SPHERICAL BALLOON'S RADIUS GROWS FROM 7 CM TO 14 CM. FIND THE RATIO OF SURFACE AREAS.

$4\pi(7)^2 : 4\pi(14)^2 = 49 : 196 =$ **1 : 4**.

SCALING

If every length is multiplied by k, areas multiply by k^2 and volumes by k^3.

USING H FOR A CONE'S SURFACE

The curved surface uses the **slant height l**, while the volume uses the **vertical height h**.

Quick recap

- Cone: CSA = πrl, TSA = $\pi r(l + r)$, V = $\frac{1}{3}\pi r^2 h$, $l^2 = r^2 + h^2$.
- Sphere: $4\pi r^2$, $\frac{4}{3}\pi r^3$. Hemisphere: $2\pi r^2$ (curved), $3\pi r^2$ (total), $\frac{2}{3}\pi r^3$.

Practice questions

1. The slant height of a cone with radius 5 cm and height 12 cm is
   (a) 17 cm (b) 13 cm (c) 7 cm (d) 15 cm
2. The surface area of a sphere of radius 7 cm is ($\pi = 22/7$)
   (a) 154 cm^2 (b) 616 cm^2 (c) 1437 cm^2 (d) 308 cm^2
3. The total surface area of a hemisphere of radius r is
   (a) $2\pi r^2$ (b) $3\pi r^2$ (c) $4\pi r^2$ (d) πr^2
4. If the radius of a sphere is doubled, its volume becomes
   (a) 2 times (b) 4 times (c) 8 times (d) 6 times
5. The volume of a cone is what fraction of a cylinder with the same base and height?
   (a) 1/3 (b) 1/2 (c) 2/3 (d) 3

Answers: 1. b 2. b 3. b 4. c 5. a